Intent agent (LLM / heuristic / replay)
Orchestrator (Python)
Execution layer (bridge gadget, synced)
Network + demand layers

Decision epoch t (frame f_t = 1800 t)

pause simulation at f_t
build state o_t (coverage, cells, loads, disconnected positions, threats)
o_t (JSON)
o_t tagged with f_t
format observation g(o_t) (full / no-spatial / minimal)
a_t + reasoning text
parse, log latency, parse status, raw output
"f_t | a_t"
accept only if tag == f_t (stale actions ignored)
resume; schedule a_t for frame f_t + 45 (1.5 s)

Frame f_t + 45

validate a_t (terrain, water, threat zone, format)

alt **[valid]**
execute (order engineer / reconfigure slice)
log "executed", frame lag

[invalid]
log "skipped: reason" (treated as noop)

Simulation runs until f_{t+1}

economy expands, units move, civilian AI orders (deferred 45 frames)
connectivity, capacity, QoS effects on demand

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